

COURSE TITLE : CHEMICAL CAD LAB
COURSE CODE : 4077
COURSE CATEGORY : A
PERIODS/ WEEK : 6
PERIODS/ SEMESTER : 84
CREDIT : 3

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Piping layout with 2D	22
2	2D Modeling of Heat Exchangers & Evaporators	25
3	3D Modeling of Driers, Extractors & Crystallizers	20
4	Process Instrumentation diagrams	17
TOTAL		84

COURSE OUTCOME

SL.NO.	GO	STUDENT WILL BE ABLE TO
1	1	Set up 2D piping layouts.
	2	Setup isometric piping layouts.
	3	Set up drawings of various flow meters.
2	1	Carry out drawings of different types of Heat exchangers.
	2	Carry out drawings of different types of evaporators.
	3	Demonstrate working of Heat Exchangers and Evaporators.
3	1	Identify basic types of 3D modeling.
	2	Set up 3D models of Driers
	3	Set up 3D models of Extractors
	4	Set up 3D models of Crystallizers.
4	1	Set up process instrumentation diagrams.
	2	Set up process instrumentation diagrams in 3D.
	3	Demonstrate process instrumentation with 3D models.

CONTENT DETAILS

MODULE – I

Piping layout with 2D

Piping layout with pipe joints, valves elbows, pumps etc. Exercise in drawing (Schematic) single line diagram of pipe layouts using standard conventions –simple piping lay out exercises in 2D and isometric view. Line diagrams of the following – Manometers – U – Tube- Differential – inclined, etc – Orifice meter – venturimeter, Rotameter, Wet gas meter – Magnetic Flow meter.

MODULE – II

Draw Heat Exchangers & Evaporators

Fixed tube sheet heat exchanger, floating head heat exchanger, U-tube exchanger, 2-4- floating heat exchanger. Long tube vertical evaporator, Horizontal tube evaporator, forced circulation evaporator, – multiple effect evaporators – forward feed, back ward feed, mixed feed, parallel feed.

MODULE III

3D Modelling

Basic types of modelling (wireframe, surface and solid modelling)

View Command – VPOINT command – VIEWPORTS (VPORTS). UNION and SUBTRACT COMMANDS – MASSPROPERTY.

Chemical Engineering Equipment Drawing With 3D

Driers, Extractors & Crystallisers

Rotary drier, Spray drier, Boll man extractor and Agitated batch crystallizer.

MODULE- IV

Process instrumentation diagram

Heat exchanger, Dryer, Reaction vessel and Distillation column.

REFERENCES

- | | | | |
|------------------|--------------------------------|------------------------------------|----------------|
| Cheryl R. Shrock | - Advanced AutoCAD2013 | - New Age international Publishers | - First |
| | Exercise workbook for Windows | | |
| K. Venugopal | - Engineering Drawing and | - New Age international Publishers | - Fourth |
| | Graphics + AutoCAD | | |
| P. Radhakrishnan | | | |
| S. Subramanyan | - CAD/CAM/CIM | - New Age international Publishers | - Third |
| V. Raju | | | |
| P I. Varghese | - Machine Drawing | - VIP Publishers | - Twenty Sixth |
| Don W Green | | | |
| Robert Perry | - Perry's Chemical Engineering | - Mc Graw-Hill | - Eighth |
| | Hand Book | | |